

SS 681:2022
(ICS 97.220.01; 97.220.10)

SINGAPORE STANDARD

Code of practice for sport safety



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Foreword

This Singapore Standard was prepared by the Working Group on Sport Safety set up by the Technical Committee on Workplace Health and Safety under the purview of the Quality and Safety Standards Committee.

This Singapore Standard is based on Sport Singapore's 2019 Sports Safety Committee report.

This standard provides requirements and recommendations to improve sport safety. It covers a wide range of settings, including competitive and school sport, recreation and fitness activities, and addresses the risks and challenges to good safety practice that were encountered in recent years. It advocates a strong proactive approach to managing risks, especially with the desire to increase safe sport participation across all demographics, with the aim to foster a culture of continuous improvement to safety that is both prudent and practical in application.

This standard also places emphasis on the roles and responsibilities of individual participants, athletes, coaches, sport officials, volunteers, educational institutions and event organisers in various aspects of sport safety. It sets out key responsibilities to be assigned to specific personnel in risk monitoring and review, communication and training of other team members on risk management, and will better equip sport facilities to identify the sources of risks.

In preparing this standard, reference was made to TR 71 : 2019, "Code of practice for water safety – Aquatic facilities".

Acknowledgement is made to Sport Singapore for their kind permissions to use the materials from 2019 Sports Safety Committee Report for the development of this standard.

Attention is drawn to the possibility that some of the elements of this Singapore Standard may be the subject of patent rights. Enterprise Singapore shall not be held responsible for identifying any or all of such patent rights.

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0 Introduction

0.1 With rising interest and changing trends in sport due to increased participation in extreme, ultra-endurance, combative and mass-participation sport, it is important to enhance safety levels. Exercise promotion is integral to the management of various chronic diseases. This standard outlines requirements and recommendations relevant to the present landscape of sport.

0.2 With a systematic approach to the various aspects involved in sport safety, event organisers and relevant associations should work together with key stakeholders to implement the recommendations outlined in this standard to eliminate preventable incidents and take a zero-tolerance approach on accidents leading to deaths and injuries.

0.3 This standard aims to raise sport safety standards through a strong preventative approach to manage risks. The sporting communities should implement risk management protocols and a robust medical support plan as detailed in this standard.

1 Scope

This standard is applicable across a range of settings including competitive sport, school sport, active recreation and fitness activities. While it aims to cover a spectrum of sport activities adequately, it does not provide in-depth coverage of highly specific areas relating to certain sports.

The focus areas are:

- (a) fundamental principles of sport safety;
- (b) training and education in sport safety;
- (c) water safety;
- (d) sudden cardiac arrest (SCA) and death in sport;
- (e) pre-participation screening;
- (f) heat injuries in sport;
- (g) event medical support plan;
- (h) exercise and training facilities; and
- (i) surveillance, evaluation and follow-up.

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2 Normative references

The following referenced documents are indispensable for the application of this standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

TR 71	Code of practice for water safety – Aquatic facilities
SS 556	Code of practice for the design and management of aquatic facilities

NOTE – TR 71 is under revision.

3 Terms and definitions

For the purpose of this Singapore Standard, the following terms and definitions apply.

3.1 Aquatic facility

An artificial body of water used for sport, recreation, or therapeutic water activities.

3.2 Cardiorespiratory fitness

Cardiorespiratory fitness refers to the ability of the circulatory and respiratory systems to supply oxygen to skeletal muscles during sustained physical activity.

3.3 Club

An organisation of people, with a common purpose or interest, who meet regularly and take part in shared activities.

3.4 Coronary artery disease

Coronary artery disease is the build-up of plaque in the arteries that supply oxygen-rich blood to your heart.

3.5 Detraining

Partial or complete loss of training-induced adaptation in response to an insufficient training stimulus.

3.6 Educational institutions

Private and public schools that may include pre-school, childcare, primary, secondary, technical trade or vocational schools, polytechnics, junior college, college, and universities.

3.7 Event organiser

The representative(s) of organisation(s) in charge of, manages, oversees an event either fully or substantially.

3.8 Exercise and training facilities

Organisations that offer health and fitness programmes as their primary or secondary service, or promote high-intensity recreational physical activity (e.g. basketball, tennis, racquetball and swim clubs). Facilities that only provide space and equipment (e.g. unsupervised hotel exercise rooms) are also included.

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3.9 Facility operator

Person(s) who has overall responsibility for the management of the facility.

3.10 Facility owner

Facility owner includes an owner, a charterer or a lessee of a facility.

3.11 Heatstroke

Heatstroke is the most hazardous condition in a spectrum of illnesses progressing from heat exhaustion to heatstroke, in which a shared finding is hyperthermia (i.e., the rise in core body temperature when heat accumulation overrides heat dissipation during exercise or exposure to environmental heat stress). Clinically, heatstroke is characterised by central nervous system, dysfunction, multiorgan failure, and extreme hyperthermia (usually more than 40.5 °C).

3.12 Hypertrophic cardiomyopathy

Heart condition with abnormally thickened heart muscles.

3.13 Hyponatremia

Hyponatremia means that the sodium level in the blood is below normal.

3.14 Long QT syndrome

Long QT syndrome is a heart condition characterised by abnormal heart rhythm, potentially leading to sudden death.

3.15 Medical personnel

Medical personnel can range from registered healthcare professionals to non-healthcare personnel who are appointed into medical or medically related roles.

3.16 Open water

Includes beach fronts, canals, rivers, lakes and reservoirs.

3.17 Open water activities

Open water activities can be classified into the following five categories:

- (a) Recreational activities;
- (b) Competitions;
- (c) Underwater activities;
- (d) Water sporting events; and
- (e) Open water swimming lessons.

3.18 Participant

Individual who takes part in or become involved in a particular activity.

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3.19 Pre-participation screening

Pre-participation screening ranges from self-administered questionnaires to full medical screenings conducted by medical specialists. The process can start from the basic end, i.e. with a self-administered questionnaire and progressively escalate to a full and in-depth health screening by medical specialists.

NOTE – Examples of pre-participation screenings include the following:

- (a) An individual fills up the questionnaire which flags the need to see a doctor, then patient consults a general practitioner (GP) who performs some basic examination that reveals further evaluation is needed, and then the GP refers to a specialist for specific tests to evaluate the abnormalities picked up by the GP).
- (b) The patient can skip the earlier stages in (a) and go directly to a specialist for a comprehensive screening test battery.

3.20 Sport organisation

Any organisation that organises and/or manages activities in the area of sport and development, or uses sport as a tool to reach development goals.

3.21 Sudden cardiac arrest (SCA)/Sudden cardiac death (SCD)

SCA/SCD is unexpected death due to cardiac causes that usually occurs instantaneously or within one hour of physical activity.

3.22 Use of “shall”, “should”, “may” and “can”

In this standard, the following verbal forms are used:

- “shall” indicates that the requirement is strictly to be followed in order to conform to the standard and from which no deviation is permitted.
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capacity.

4 Fundamental principles of sport safety

4.1 General

4.1.1 Sport organisations and sport event organisers should have a sound and proactive safety management system in place. Implementing and maintaining a safety management plan fosters a culture of continuous improvement.

4.1.2 Sport organisations and sport event organisers shall conduct risk assessments to identify risk sources as well as the approaches to mitigate risks.

4.1.3 Risk management planning includes a systematic application of management policies, procedures and practices. It also includes tasks of establishing the context, identifying, evaluating, managing, monitoring and communicating risks. It is important to identify the root of the issue and work through each step of the process. This prevents implementing “quick-fixes” that can result in problems.

NOTE – Resources on risk assessment and management systems are available on the Sport Singapore website. [\[5\]](#)

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4.1.4 Risk assessment is an integral component of a safety management system. With effective risk management, the potential costs and liabilities of sport incidents can be minimised, thereby enabling a safer sporting experience. Sport and recreational activities involve physical risks, which vary with the type and timing of the activity, location, participants' skill level and the number of spectators. Risks are managed through assessing their likelihood of occurrence and potential consequences, developing clear actions and designing a response plan. Legal compliance and insurance coverage should also be considered.

4.2 Safety management system

4.2.1 A safety management system should be benchmarked against best practices. A sound and sustainable safety management system with effective implementation and continual upgrading is critical in the development of an incident-free sport safety culture. With effective implementation of a rigorous system, a sound sport safety culture can be established. This is the key to sustaining high standards of sport safety in Singapore. A proactive approach instead of reactionary measures whenever events or accidents occur should be adopted. A long accident-free period can result in complacency, and safety measures can be overlooked or neglected. Unsafe attitudes, behaviours and acts cause most serious incidents rather than unsafe equipment or conditions. Thus, a safety management system ensures every personnel remains vigilant and prepared.

4.2.2 A sport safety culture should involve every stakeholder including participants, event organisers, sport organisations, coaches, teachers, instructors, parents and members of the public. It is important to cultivate the awareness that everyone has a part to play.

4.2.3 Sport participants shall be personally responsible to take measures for their own safety. Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should take measures to manage safety at sport events, training programmes and facilities.

4.2.4 All in all, effort should be made to assess the risks associated with sport events and activities, and preventive measures taken to mitigate adverse events through a holistic application of education, systems thinking, medical support and operational planning.

4.3 Principles of sport safety

4.3.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should utilise a safety framework to guide the management of their plans.

A good framework should cover, but not be limited to the following areas:

- (a) Management commitment for safety, including the allocation of adequate resources;
- (b) Safety organisation and structure;
- (c) Safety attitudes, policies and principles;
- (d) Goals and objectives;
- (e) Setting high standards for sustainable performance;
- (f) Clear roles and responsibilities;
- (g) Effective communication of plans and measures;

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- (h) Equipment deployment, regular inspections and maintenance;
- (i) Emergency response and planning;
- (j) Risk assessment and management;
- (k) Training and development, including regular safety drills reflecting a variety of scenarios;
- (l) Supporting safety personnel;
- (m) Management of changes (people, equipment, technology);
- (n) Pre-event, intra-event and post-event safety measures;
- (o) Incident reporting and documentation, investigation, evaluation, and corrective action;
- (p) Pre-event and post-event safety review; and
- (q) Regular review of the safety management system.

4.3.2 Risk management frameworks are useful in the planning process. These frameworks are by no means exhaustive or all-inclusive, and serve mainly to show the possible outlines of risk management plans.

4.3.3 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors shall make risk management an ongoing process to ensure safety is well managed, support the achievement of safety goals, and minimise the likelihood of risks.

The risk management process involves:

- (a) risk assessment;
- (b) risk mitigation;
- (c) communication of risks; and
- (d) review and monitoring of the risk assessment.

NOTE – Resources on risk assessment methodology/template can be downloaded from the Sport Singapore website. [\[6\]](#)

4.3.4 Key responsibilities should be clearly assigned to specific people in areas such as risk monitoring and review, communication and training of team members on risk management. Everyone involved should be informed and aware of the risks to be managed

4.3.5 Risk management documentation should include:

- (a) a comprehensive risk management plan;
- (b) an incident/accident registry; and
- (c) a post event review/report.

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4.3.6 Sport organisations, associations, clubs, facility owners and operators, event organisers, coaches and instructors shall conduct risk assessments to allow the sources of risks to be identified and reasonably practicable steps taken to eliminate any foreseeable risk to those involved. Where it is not possible to eliminate risk, other reasonably practicable measures shall be taken to minimise risk.

4.3.7 Sport organisations, associations, clubs, facility owners and operators, event organisers, coaches and instructors shall conduct risk assessment by addressing the following:

- (a) Unexpected or negative things that could happen;
- (b) Likely impact and mitigating actions; and
- (c) Keeping everyone involved informed.

4.3.8 When identifying risks, the following factors should be considered:

- (a) Age and capacity of participants, especially those who require extra care (e.g. young children, the elderly, those with special needs);
- (b) Types of activities;
- (c) Personal history of accidents, injuries and losses;
- (d) Rescue and lifesaving equipment;
- (e) Sport equipment;
- (f) Current and potential issues with standard operating procedures or practices;
- (g) Facilities and their infrastructure; and
- (h) Environment where the activity is happening.

5 Safety training and education

5.1 General

5.1.1 There has been a rapid growth in sport participation rates and a large population of novices are taking part in events. Novices may have limited knowledge regarding the risks involved in their sport/activities making them susceptible to injury. Being inexperienced, they may not be aware of their physical limits. Apart from changes in the mix of participants, there may be the addition of new events as sport evolves (e.g. the addition of kiteboarding), presenting “uncharted territory” and new challenges in terms of safety measures.

5.1.2 Sport safety is not only about ensuring that each sporting activity is being conducted in a safe environment but includes many facets such as risk assessment, pre-event physical and mental health, and the ability to act immediately during accidents or emergencies. Sport safety is a shared responsibility and everyone from coaches, technical officials to management has a part to play.

5.1.3 Training and education in sport safety should address the various facets of sport safety and be specific and tailored to the application context. Sport organisers, coaches, instructors, volunteers, sport participants and their family members can benefit from the training and education. In an emergency, they can act as first responders to provide assistance at the scene.

NOTE – Sport organisers, national sport associations, sport clubs, educational institutions, parents, and anyone who employ coaches are encouraged to only engage those that are registered by a credible organisation, e.g. the National Registry of Coaches.

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5.1.4 A sport safety awareness programme for all stakeholders should help to nurture Singapore's sport safety culture.

5.1.5 For a sport safety culture to take root, it should have a multi-pronged approach targeting sport organisations, associations, clubs, event organisers, coaches, instructors, facility owners and operators, volunteers, sport participants and their families.

The programme should:

- (a) raise awareness and knowledge of sport safety in all the above groups;
- (b) increase accessibility to sport safety training, including life-sustaining first aid; and
- (c) promote lifelong learning in sport safety through refresher training programmes.

5.1.6 Sport organisations, associations, clubs, event organisers, coaches, instructors, facility owners and operators, and educational institutions should provide educational efforts and resources to ensure that athletes and sport participants know the signs and symptoms of SCD/SCA. It is of utmost importance that athletes and sport participants are educated on the important symptoms to look out for. These include exertional chest pain/discomfort, fainting or near fainting spells, palpitations, as well as excessive exertional and unexplained breathlessness/fatigue associated with exercise. They should also incorporate emerging issues into their safety training and development such as harassment and abuse, and mental health.

5.2 Sport safety programmes

5.2.1 Sport safety training programmes and resource packages should continually be enhanced and made accessible to all sport enthusiasts, the community and organisations, and cover (but not limited to) the following areas of sport safety:

- (a) Benefits of sport;
- (b) Potential dangers of unsafe sport;
- (c) Participant preparation that can aid safe participation in sport;
- (d) Safety considerations in setting up and conducting safe sporting activities;
- (e) Immediate actions that need to be taken if injuries occur during sporting activities (including the need to report early warning symptoms to a doctor during sporting activities);
- (f) Need for all stakeholders involved in sporting activities to be trained and ready to actively assist in the event of unexpected injury;
- (g) Cardiopulmonary resuscitation (CPR)/automated external defibrillator (AED) training and/or first aid training; and
- (h) Use of lifesaving equipment (e.g. AEDs, first aid kits and other medical equipment).

5.2.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors, should play a significant role in promoting sport safety in Singapore by encouraging training in sport safety awareness and promote sport safety amongst their officials, staff, athletes and participants. Sport organisers, national sport associations, sport clubs, educational institutions and the general public should engage sport coaches that are registered under a credible organisation so that sport safety standards may be applied more consistently.

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5.2.3 An environment of sport safety is everyone's responsibility. Sport participants, enthusiasts, parents, volunteers and the community should be equipped with either standard first aid and AED, CPR and AED, or at least, the citizen first responder programme (CFRP). Sport organisations, associations, clubs, facility owners and operators, educational institutions, coaches, instructors should make sport safety a priority in their annual planning and operations.

5.2.4 The families of sport participants should be encouraged to be equipped in sport safety. An outreach should be done through conducting or promoting available sport awareness and training sessions in either standard first aid, CPR and AED or, at least, CFRP as best practice standards.

6 Water safety

6.1 General

6.1.1 Drowning occurs for many different reasons, and is often due to a lack of:

- (a) awareness of dangers when engaging in open water activities (e.g. kayaking during spring tides or swimming in open water areas with strong currents);
- (b) CPR and lifesaving skills among caregivers and supervisors;
- (c) general and water safety knowledge;
- (d) swimming proficiency; and
- (e) vigilance and supervision from parents and caregivers.

6.1.2 All swimming pool users, water sport participants and enthusiasts shall practise personal responsibility to be water safe.

6.1.3 Swimming and water survival programmes as well as water safety awareness programmes are necessary for public education.

6.1.4 Adults shall exercise care and responsibility by abiding with safety regulations and maintaining close supervision of their children.

6.1.5 Parents/caregivers shall stay attentive and within arm's reach of children when bringing them to water bodies. This enables adults to pull the children out of the water in the event of an emergency. During swimming lessons under the supervision of coaches, parents/caregivers should help with supervising their children by staying close and in direct line of sight of the children.

NOTE – Refer to the Sport Singapore website for "Coach's Code of Ethics" [\[7\]](#).

6.1.6 Aquatic and water sport facilities should support surveillance efforts and provide rescue equipment such as rescue tubes, poles and throw ropes with instructions.

6.1.7 Not all swimming pools, beaches and recreational water bodies are staffed with lifeguards. Operators and event organisers shall have in place mitigating measures to proactively prevent drowning incidents.

6.1.8 The nature and conditions of open water environments shall be well understood by users so that they are in a better position to assess the risks.

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6.1.9 Individuals shall take note of (but not limited to) the following environmental hazards:

- (a) Water temperature;
- (b) Water currents and eddies, including riptides;
- (c) Tidal flows;
- (d) Water quality;
- (e) Obscure underwater and overhanging hazards;
- (f) Water depth;
- (g) Conditions underfoot;
- (h) Marine animals;
- (i) Artificial structures like floating barrels;
- (j) Weather conditions (e.g. haze); and
- (k) Ultraviolet (UV) rays.

6.1.10 Event organisers shall identify the roles and responsibilities of people involved in each activity.

6.1.11 Individuals shall be responsible for their own safety and take necessary preventive measures such as using a pre-participation questionnaire, personal floatation devices if necessary, wearing appropriate attire, and protecting themselves from the environmental hazards mentioned above.

6.1.12 The higher risks of open water shall require facility owners and operators, event organisers, users, volunteers, lifeguards, organising and instructors to understand the safety requirements and practices. Safety and rescue equipment shall be made available. A risk assessment shall be conducted to identify risks and determine the appropriate control measures.

6.2 Safe practices for aquatic facilities

6.2.1 Swimming and water survival programmes are essential for water safety. Individual and facility owners and operators should promote swimming proficiency programmes and water safety training.

6.2.2 Water safety awareness programmes should be organised by facility owners and operators (including educational institutions) to advocate key areas of action covering (but not limited to) the following topics:

- (a) CPR / AED usage;
- (b) Drowning and near-drowning case studies;
- (c) Lifesaving and rescue equipment;
- (d) Pre-participation questionnaire usage;
- (e) Prevention of shallow water blackouts;
- (f) Stretching and warm-up exercises;

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- (g) Swimmers' behaviours and identification of swimmers in distress; and
- (h) Swimming and water survival skills.

6.2.3 Infants, toddlers and pre-schoolers shall be under constant adult supervision when they are in and around water. The adult/caregiver should be familiar with the child's capabilities and equip the child with personal floatation devices, if necessary. Even with floatation devices, constant supervision of children is necessary to ensure their safety.

6.2.4 Surveillance and rescue efforts may be supported by technology. Installing surveillance systems eases the process of supervision in aquatic facilities and can possibly reduce the risk of drowning.

6.2.5 Facility owners and operators shall adopt the following water safety precautions/advisories for swimmers:

- (a) Follow safety rules and signs;
- (b) Never swim alone;
- (c) Learn swimming and water survival skills;
- (d) Understand the dangers of water;
- (e) Supervise children and weak swimmers at all times; and
- (f) Swim in safe areas.

6.2.6 Hotels and condominiums with aquatic facilities, sport event organisers and communities using aquatic facilities shall refer to TR 71 and adopt the relevant clauses.

6.3 Safe practices for open water

6.3.1 Individuals shall be responsible for their own safety and need to understand the nature and conditions of open water.

The following shall apply to individuals and participants when engaging in open water activities:

- (a) Seek information on the safety of beaches or water bodies, as well as any deployment of trained lifeguards on duty. It is essential to abide by the safety regulations and guidelines of the facility owner/operator/event organiser.
- (b) Exercise within their capabilities and use personal floatation devices where necessary.
- (c) Avoid consumption of alcohol or any medication that can cause drowsiness or impair judgement before participation. However, medication for chronic diseases should still be taken.
- (d) Close supervision of children by a competent adult/caregiver.

6.3.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors shall conduct mandatory safety briefings prior to the start of the open water event/activities (e.g. kayaking or stand up paddle board expeditions).

6.3.3 Open water locations shall have physical structures to inform users of critical safety information. Physical structures such as information signs and pontoons are useful in conveying critical safety information. This includes restricted swim zones, restricted water activities, procedures of using the safety equipment, etc.

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6.3.4 A public announcement system should disseminate critical information when required. Warning sirens should be used to inform users of possible emergency or dangers and the activation of the siren can be used during inclement weather, change of tides, release of water, or when dangerous activities are spotted.

6.3.5 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should consider using technology as it can play a crucial role in improving the safety of open water users through (but not limited to) the following means:

- (a) Communication devices – aids efficient communication between safety staff;
- (b) Security alarms – to be used within authorised premises;
- (c) Emergency alarms – for audio warnings during inclement weather, change of water flow, opening of tidal gates at reservoirs, dangerous occurrences, etc;
- (d) Closed circuit television – for wider coverage and more efficient monitoring by lifeguards/security officers; and
- (e) Surveillance systems – use of technology such as, but not limited to, a drone to alert personnel of victims in distress.

7 Sudden cardiac arrest / Sudden cardiac death (SCA/SCD)

7.1 General

7.1.1 Regular physical activity confers numerous health benefits for individuals of all ages. The increase in cardiorespiratory fitness through exercise has also shown to reduce the risk of all-cause mortality. However, there is a paradoxical increase in the risk of SCA and/or SCD during or shortly after exercise, particularly during vigorous exercise. However, the risk of SCD and sport related injuries while participating in sport is low and the benefits of regular exercise outweigh these risks.

7.1.2 Sport-related SCA/SCD in athletes under 35 years of age is usually associated with underlying structural, electrical or biochemical heart abnormalities. The major cause of SCA in athletes above 35 years of age is coronary artery disease. It is likely that SCA/SCD risk varies with age, gender, intensity of activity, race and ethnicity.

7.1.3 Athletes who fit the risk profile should undergo cardiac screening by a doctor. When found to have an underlying heart disease, they should consult a doctor for advice before embarking on any physical activity.

7.1.4 AED coverage shall be provided for sport which has high cardiovascular intensity and/or participants at risk especially elderly and people with high risk of heart disease.

7.2 Recognition of warning symptoms

7.2.1 Warning symptoms may precede SCA. Recreational and competitive athletes should be educated on the important early warning symptoms and signs to look out for, and to seek further medical review to evaluate these symptoms. These include exertional chest pain/discomfort, fainting or near fainting spells, palpitations, as well as excessive exertional and unexplained breathlessness/fatigue associated with exercise. If detected, athletes should undergo further medical reviews to evaluate these symptoms.

7.2.2 Athletes should also be educated on measures to exercise safely such as refraining from exercise when they have a fever or other systemic signs of infection.

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7.2.3 Athletes (under 50 years old) with positive family history of premature heart diseases or cardiac death, or hereditary cardiac conditions should seek cardiac clearance for sporting participation and competition. This includes those with a family history of specific hereditary heart condition (e.g. hypertrophic cardiomyopathy or long QT syndrome).

7.2.4 Sport organisations, associations, clubs, educational institutions and event organisers should support with educational efforts and resources to provide information on the signs and symptoms, as well as reporting channels that athletes and participants can openly relate to. It is of utmost importance that they are educated on the important early warning symptoms and signs to look out for that require medical attention before embarking on sporting activities or exercise. Wide-reaching public education is critical in raising public awareness on issues in sport safety. These include indications for seeking medical review, limitations of screening and benefits of exercise versus risk of SCD from exercise.

7.3 Training

7.3.1 Foreign athletes from temperate climates should be encouraged to spend some time acclimatising to local climate and humidity prior to the competition.

7.3.2 To minimise the risk of SCA, athletes should have adequate and appropriate training with a gradual build-up in volume and intensity. Exercise capacity drops with detraining, thus for periods in which exercise routines may be halted (e.g. owing to illness or work), athletes should resume training at a level lower than when they stopped.

7.3.3 Cardiac collapses in a marathon often occur at the final mile or finishing line. During the race, athletes should run at a steady pace and avoid sprinting in the last mile of the race unless there is sufficient specific conditioning. They should also not push themselves towards the end of the race especially without adequate preparatory training.

8 Pre-participation screening

8.1 Pre-participation screening principles

8.1.1 People of all ages and levels of fitness engage in sporting activities of low, moderate or high intensity. Every year, a number of individuals of various age groups sustain cardiovascular collapse while engaged in active sporting activity. The numbers who develop SCA during sporting activities are small. Pre-participation screening aims to pick up existing conditions that may cause harm during strenuous physical activities.

8.1.2 A system of pre-participation screening is useful in predicting a person's likelihood of having a cardiac arrest. Through careful history and evaluation, the factors contributing to cardiac arrest (i.e. drugs and infections) can be potentially deduced.

8.1.3 The risk of any physical activity is an interaction of the exercise per se and the individual's fitness and medical conditions. The competitiveness of the individual should be considered as it can increase the risk, especially in an overzealous participant/competitor.

8.1.4 Individuals shall complete pre-participation questionnaire prior to embarking on strenuous physical activities. Pre-participation screening aims to pick up existing conditions that may cause harm during strenuous physical activities.

NOTE – Refer to the Sport Singapore website for an example of a pre-participation questionnaire [\[8\]](#).

8.1.5 For the selection of the appropriate type of screening, it is based on the competition level, type of sport and individual's risk level minimally.

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8.1.6 Individuals shall routinely complete, at the least, the pre-participation questionnaire prior to joining a club (e.g. gym), competition, course, or organised sport activity. It shall be completed before the commencement of the physical activity. The questionnaires identify individuals with known conditions as well as those without any prior history of medical illness but who have symptoms or a history of medical conditions such as chest pain, breathlessness, fainting, dizziness, or palpitations. If indicated by the questionnaire, the participant should consult a doctor.

8.1.7 The pre-participation questionnaire shall be completed minimally on an annual basis as new symptoms may develop after the completion of the previous questionnaire. All individuals involved in sport shall take personal responsibility for their own health, and make use of these simple-to-use and self-administered pre-participation questionnaire. Within a year, individuals can also choose to complete pre-participation questionnaire as and when required; regardless of the sport activity.

8.1.8 Athletes and coaches should be educated on the pre-participation questionnaire to help raise awareness.

8.2 Use of pre-participation screening

8.2.1 Sport organisations, associations, clubs, facility owners and operators, event organisers, coaches and instructors should have some form of pre-participation screening to identify the cohort of athletes and sport enthusiasts affected by unsuspected underlying cardiovascular disease. This enables appropriate interventions to prevent SCD during sporting activities.

8.2.2 Individuals who are involved in sporting activities shall take some measures for his/her own safety by actively participating in safety screening procedures. This will best inculcate the culture of sport safety that is necessary for minimising the prevalence of injuries during sporting activities.

8.2.3 Pre-participation screening may start, as a minimum standard, with the use of a self-administered questionnaire prior to all sporting activities which identifies individuals who require further evaluation by a doctor. After stratifying the risks, the doctor, at his or her clinical discretion, shall order further screening of the identified at-risk individuals based on the following:

- (a) Individual's intrinsic risk of sudden death or serious injuries (e.g. prodromal symptoms, positive family history);
- (b) Current health condition;
- (c) Level of competition in accordance with Figure 1; and
- (d) Degree of risk of the particular sport or activity (refer to Table 1 and Table 2).

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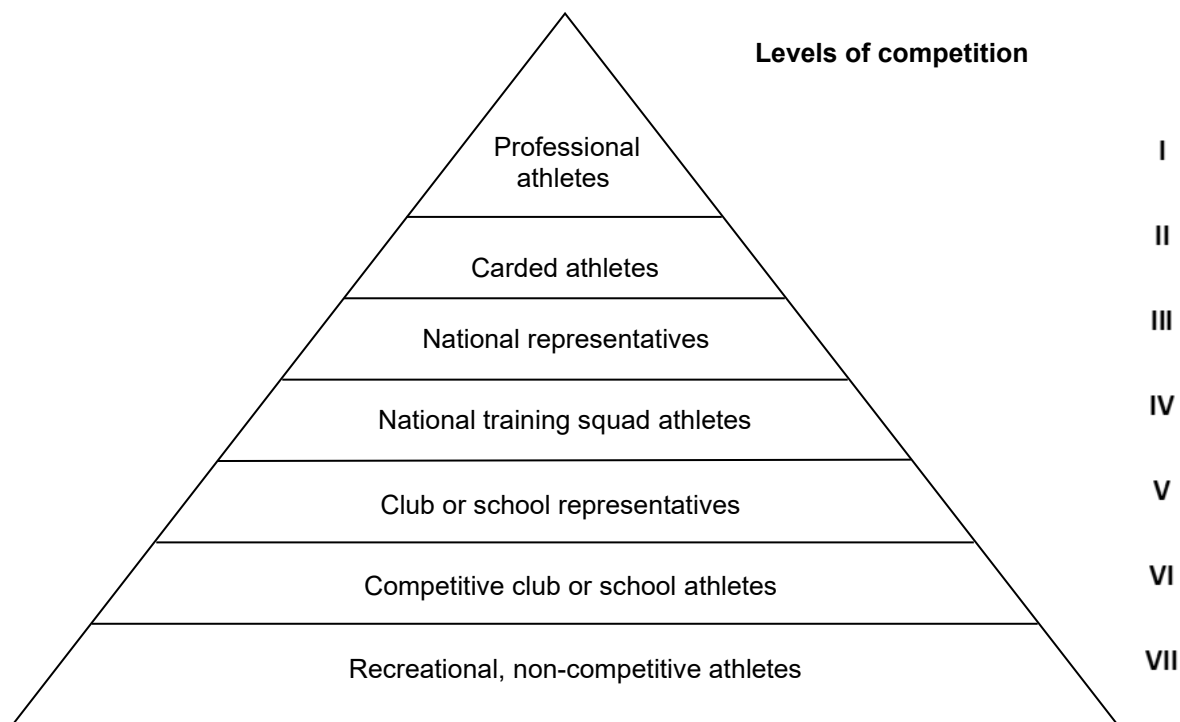


Figure 1 – Athletes’ competitive levels

Table 1 – Categorisation of event risk

Category	A	B	C
Risk	Low	Medium	High

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Table 2 – Classification of sports based on cardiovascular activity and contact/collision risk

		Contact/collision risk		
		Low	Medium	High
Cardiovascular risk	High	Canoeing/kayaking Dragon boat rowing	Cycling Ice skating (speed) Roller sports	Boxing/wrestling Muay Thai Mixed martial arts
	High moderate	Running (mid-distance) Bodybuilding Swimming (pool) Modern pentathlon	Badminton Squash Triathlon Open water swimming	Basketball Handball Ice hockey Extreme sports
	Moderate	Running (spring) Running (long-distance) Field (throwing) DanceSport Mountaineering Pickleball Sailing Lifesaving programme Tennis	Field (jumping) Floorball Ice skating (figure) Water ski/wakeboard Gymnastics Skiing and snowboarding Wushu Weightlifting	Soccer Hockey Judo Karate-do Kendo Rugby Silat Taekwondo Ultimate frisbee

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		Contact/collision risk		
		Low	Medium	High
Cardiovascular risk	Low moderate	Archery Table tennis Underwater activities ¹ Scuba	Fencing Netball Sepak takraw Baseball/softball Cricket Volleyball Platform diving	Cheerleading Equestrian Motor sports Powerboat
	Low	Bowling/bowls Chess Contract bridge Cuesports Darts Gateball/woodball Golf Lawn ball Pentanque Shooting Weiqi/Xiangqi		
Key ¹ Underwater activities carry a different set of risks such as drowning, hyperbaric injuries. An activity-specific risk assessment should be performed by the event organising committee.				

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8.2.4 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should minimise the hindrance to sport participation and sport excellence when administrating the pre-participating screening.

8.2.5 Individuals should be educated on symptoms and signs that require medical attention before embarking on sporting activities or exercise. The recognition that pre-participation screening is only part of the strategy to decrease the chance of sudden death and adverse events. Education is the other crucial component of the overall strategy.

8.2.6 Individuals participating in sport shall ensure that they undergo pre-participation screening annually.

8.2.7 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should cultivate a sport safety culture through awareness programmes and conducting pre-participation screening annually.

8.2.8 Individuals who encounter symptoms, such as palpitations, fainting spells or seizures or dizziness or chest pain or unexplained breathlessness, extreme fatigue during exercise or other sporting activities should see a medical practitioner as early as possible and have these carefully evaluated. This is in view of the presence of prior early warning symptoms in a number of persons who have suffered SCA during sporting activities. Education on the need for this should be included in sport safety awareness briefings and the physical education (PE) curricula in educational institutions. Medical practitioners should also carefully evaluate these symptoms with a view to excluding underlying heart disease that could be potentially life-threatening.

8.3 Risk stratification

8.3.1 Risk stratification based on participant's profile, competitive intensity and type of sport for physician-led screening and competitive levels should be used in consideration with other risk factors such as the type of sport, state of training and other inherent risk factors. Athletes can be risk-stratified based on the level of competition (refer to Figure 1). Generally, with a higher level of competition comes a higher training intensity and volume, and therefore, a higher risk of sudden death and injuries.

8.3.2 Cardiovascular activity should be chosen as the main factor owing to its stronger association with known intrinsic risk factors. The duration of participation, contact/collision risk, or environmental stress, can also affect risk levels. Sport organisations, associations, clubs, facility owners and operators, event organisers, coaches and instructors should exercise discretion with the sport-specific guidance of their medical advisors, medical committees, or international federations, where applicable.

8.3.3 The practice of compulsory annual pre-participation screening for professional athletes (level I) and carded athletes (level II) (refer to Figure 1), regardless of sport category, should be continued.

8.3.4 Sport organisations responsible for national representatives (level III) and national training squad athletes (level IV) should ensure that they undergo pre-participation screening annually regardless of their sport category. The training intensity and volume for these athletes are very similar to that of the professional athletes (level I) and carded athletes (level II). As carding is not entirely based on training intensity and volume, it should not be used to differentiate who should or should not be screened. For example, an elite athlete training at a high intensity and volume may not be carded because his sport is not included in the major games. Another example is how the athletes of the Singapore Sports School spend more time training than a typical school athlete (or even some of those who represent Singapore) and should therefore be treated as level III and IV athletes. As such, they should continue to be required to undergo annual pre-participation screening.

8.3.5 The pre-participation screening for these categories of athletes should comprise:

- (a) Annual pre-participation screening using a self-administered questionnaire.

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- (b) Any other types of screening procedures determined necessary by the medical authority of the sport organisations, such as 12-lead Electrocardiogram (ECG) or pre-participation physical evaluation or both. All sport organisations should appoint either a formal medical committee comprising relevant expertise (sport physician or sport medical practitioner or other physician with a keen interest in the sport), or at least a medical advisor to advise the sport organisation's executive committee on medical matters, including pre-participation screening.
- (c) If abnormalities are identified during pre-participation screening, the individual should be referred to the appropriate medical facility / specialty.
- (d) All sport organisations should finalise the screening protocol required for their respective sport, and submit the protocol to the relevant national sports association(s) for endorsement and standardisation across the sport.

8.3.6 For club representatives (level V) and competitive club athletes (level VI), annual pre-participation screening (pre-participation questionnaire at the very least), is strongly recommended. As clubs, they should be able to demonstrate evidence of such screening during official competitions. Sport clubs can contribute to reasonable, common standards of sport safety in the country by implementing these recommendations within two years of the release of this report.

8.3.7 For categories B and C sport (refer to Table 2), it is recommended that event organisers should reinforce the sport safety message by reminding participants about the need to either perform pre-participation questionnaire screening or, at least, provide verbal advisory well-being checks before the start of the event. Follow-up action should be taken as indicated by the questionnaire or response to the advisories.

8.3.8 For club or school representatives (level V), competitive club or school Athletes (level VI) and recreational, non-competitive athletes (level VII), as a minimum, the pre-participation questionnaire should be completed at least annually. For this group, pre-participation screening just before each sporting event would be challenging to be made mandatory owing to the large numbers.

9 Heat injuries

9.1 General

9.1.1 Event organisers, coaches, sport participants and athletes should learn to manage heat stress when training and competing in Singapore.

9.1.2 Adherence to good practices can reduce the risk of heat injury. This involves progressive training, heat acclimation and early identification of risk factors.

9.1.3 Members of the public, amateurs and professional athletes who exercise and are not acclimatised for hot climates face higher risks of heat injury. Adherence to the following good practices may mitigate the risks of heat injury:

- (a) For the individuals, this involves progressive training and personal readiness to optimise thermoregulatory capacity; and
- (b) For coaches and event organisers, additional measures may be required to effectively manage the risks of heat injury, over and above the risks of traumatic injury and cardiac events.

9.1.4 Early recognition of symptoms of exertional heatstroke and rapid first responder intervention can reduce morbidity and mortality. Refer to 9.5.3 on recommendations to responding to exertional heatstroke.

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9.1.5 Heatstroke can also be triggered by suboptimal immune function. Immune suppression can result from the combination of prolonged intense training coupled with the lack of opportunity for recovery, subclinical infection and recent illness.

9.2 Heat injury awareness

9.2.1 Sport participants, athletes, coaches, instructors and event organisers should be educated on how to prevent heat injuries. It is critical to identify and respond to risk factors such as recent illness, and to seek medical advice prior to any sport activity. The checklist of intrinsic risk factors for exertional heatstroke is found in Annex A.

9.2.2 Sport participants, athletes, coaches, instructors and event organisers should be aware of heat injury prevention techniques such as gradual conditioning and acclimatisation to heat.

They should also note the following:

- (a) Sport participants and athletes (especially foreigners) intending to compete in a hot and humid environment should appreciate heat acclimatisation as a means of reducing heat strain.
- (b) Acclimatisation training shall be supervised by an experienced coach or with partner(s). Before the event, acclimatisation can be achieved through gradual daily exposure to heat strain over a period of 10 to 14 days in an external environment or in an environmental chamber (more commonly known as heat acclimation). With acclimatisation training, the body has enhanced responsiveness in its thermoregulatory mechanisms during exercise, as well as an increased level of work tolerance in the heat.
- (c) Sport participants, athletes and coaches should weigh the use of protective and heat-trapping clothing in heat acclimatisation against the increased risks of injury during acclimatisation training. The risk of heat injury should be considered and minimised if the heat acclimatisation programme is done with protective and heat-trapping clothing.
- (d) Sport participants, athletes and coaches may enhance competitive performance by training during the hotter hours of the day, with sufficient attention given to fluid replacement and gradual progression. The physiological adaptations that occur as in the heat acclimatisation process also normally contribute to performance enhancement.

9.2.3 Sport participants, athletes, coaches, instructors, event staff, volunteers and medical personnel should be educated on how to recognise the early signs of heat injury and how to activate an emergency medical responder when the following signs are observed:

- (a) Symptoms of heat injury (as experienced by athletes):
 - Giddiness and dizziness;
 - Headache;
 - Light-headedness, fatigue, drowsiness;
 - Nausea;
 - Numbness, cramps, weakness of extremities; or
 - Shortness of breath even at rest.
- (b) Signs of heat injury (as witnessed by onlookers):
 - Altered mental status (confusion, disorientation, combativeness);
 - Ataxia (inability to maintain balance, frequent falling);
 - Hyperventilation;
 - Loss of consciousness;
 - Seizures; or
 - Vomiting.

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9.3 Individual responsibility

9.3.1 Sport participants, athletes, coaches and instructors shall exercise individual responsibility to understand and practise basic safety precautions when training and competing in the heat. Inadequate conditioning and heat acclimatisation, recent illness, cumulative fatigue and overtraining can increase the risks of heat injury.

9.3.2 Sport participants, athletes, coaches and instructors shall know their own state of health and be conscious of the risks of exceeding their threshold for safe participation in the activity. When in doubt, they should consult a medical practitioner with relevant experience in heat injury prevention (e.g. sport physician).

9.4 Training and competitions

9.4.1 During training and competition, sport participants, athletes, coaches and instructors should take proactive steps to minimise excessive heat accumulation. Preservation of health through good dietary habit, well-regulated sleep routine and catering for recovery in between training bouts are key measures to preventing heat injury and ill health.

9.4.2 Sport participants, athletes, coaches and instructors should pay attention to early symptoms of overtraining and illness, e.g. sore throat, diarrhoea, cough, yellowish phlegm, a blocked nose (not allergy-related) and low-grade fever. They should seek medical attention, if necessary, and keep coaches informed about the change in state of health. Training plans should be adjusted to allow them to recover before resuming intense training.

9.5 Heat injury prevention and management

9.5.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should be aware of how they can prevent heat injury. The preventive primary, secondary and tertiary measures are found in Table 3.

Table 3 – Preventive measures for coaches and event organisers

Primary prevention	Sport organisations and event organisers should provide participant with advisories on the prevention of heat injuries. Coaches should be able to identify signs and symptoms of the different types of heat injuries. This is given the prevailing environmental heat stress conditions in Singapore. Sport organisations, event organisers and coaches should also have the necessary instruments available and/or means of communication to conduct on-site risk assessments prior to the start of and through to completion of sporting activities. In the absence of formal sport-specific regulations issued by international sports federations on competing in the heat, it is advisable that organisers seek input from relevant international authorities and national governing bodies. Sport organisations, event organisers and coaches should be aware of international guidelines for high impact sport such as long-distance running, marathon events and prolonged intense activity.
Secondary prevention	Sport organisations, event organisers and coaches may also choose to institute additional breaks, dedicated cooling facilities and drinking stations to facilitate intermittent cooling and fluid intake respectively. At outdoor events such as road races, there should be adequate water stations throughout the race route. Local sport organisations, event organisers should consider having water points at every 5 km during marathons in Singapore. Measures to improve

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	air exchange, encourage flow of air and cool playing environments can be considered for training and competition during the hottest periods of the day.
Tertiary prevention	When play is in session, it is also the responsibility of sport organisations, event organisers and coaches to ensure that sport participants and athletes in distress are identified in a timely manner and that remedial actions are executed safely. While the responsibility might be entrusted to onsite safety and medical personnel, it is essential that all members of staff, coaches and athletes themselves are mindful of early symptoms of heat injury and are able to communicate to organisers should they encounter an athlete in distress.

9.5.2 Medical personnel should keep abreast of latest recommendations and practices in the emergency management of exercise-associated collapse as found in Table 4.

Table 4 – Preventive measures for medical personnel

Primary prevention	Medical personnel should advise athletes to pace themselves according to their fitness level and present health condition. Those with symptoms of a compromised immune system (e.g. runny nose, cough, sore throat) should not be exercising intensely, or be advised to refrain from physical exertion if necessary.
Secondary prevention	Medical personnel should work with athletes, coaches and organisers to ensure that early detection measures are in place during training and competition. This can be in the form of education on the warning signs of heat injury or active screening of athletes at specific checkpoints.
Tertiary prevention	When a casualty is presented to a first aid post, medical personnel shall be adequately equipped to identify the casualty and rule out differential diagnoses (including hyponatremia and cardiac events). Confusion and disorientation are indicative of exertional heatstroke in an athletic setting. Medical personnel shall communicate effectively with first responders to establish the sequence of events and possible mechanisms of injury. In the initial stages, it is important to manage the competing needs of resuscitation and cooling measures concurrently. It is desirable to have a means of assessing rectal temperature for the purpose of diagnosis and monitoring during treatment. Once the casualty has been stabilised and evacuated for further care, medical personnel shall inform organisers and advise on whether play or competition should be allowed to continue. It is important to note that in endurance events, a heat casualty is rarely an isolated incident.

9.5.3 Recommendations in responding to exertional heatstroke are as follows:

- Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should be ready to respond immediately by eliminating danger, activating emergency medical services and securing airway, breathing and circulatory support when exertional heatstroke is noticed in the participant. These measures should be identified in the risk assessment and management of exertional heatstroke cases.
- The facility should also have in place the equipment, items, and plans to execute body cooling immediately when exertional heatstroke occurs and personnel shall be trained to execute these plans proficiently. Cold water immersion using ice water is most recommended for body cooling and should be the default modality for body cooling. If ice water immersion cannot be

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implemented, alternative modalities that have been proven to be effective for cooling exertional heatstroke patients include portable body cooling devices that are certified for emergency use and the combination of fanning and bathing the body with ice water soaked in towels.

- (c) Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should use ice water and cold water immersion as cooling modalities execute body cooling to achieve the highest rates of temperature reduction during heat injury. The implementation of body cooling procedures should take into consideration the need to minimise interference with other medical devices that may be used concurrently during an exertional heatstroke event.

9.5.4 The decision on return-to-play after a heat injury shall be made on a case-by-case basis depending on the nature of sport played, risk factor profile and severity of heat injury sustained.

9.6 Risks of over-hydration

9.6.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should provide sport participants and athletes with advice on the risks of over-hydration leading to water intoxication.

9.6.2 Sport participants and athletes should understand how hydration status is determined by their own behaviour before, during and after participation in moderate to vigorous physical activity. Inexperienced sport participants and athletes might face difficulties gauging and planning their own hydration needs.

9.6.3 Over-hydration occurs in athletes when the supply of fluid exceeds the body's current needs and storage capacity. Excess fluid intake is closely linked with hyponatremia or water intoxication. There is consensus that it is normal for most athletes to be dehydrated for up to 3 % of body mass loss during the activity without health implications. Dehydration in well-trained athletes can even be higher. The degree of dehydration that can be tolerated is highly individualised and sport participants and athletes should monitor their body mass changes before and after exercise to guide their hydration needs.

9.6.4 It is normal to be mildly dehydrated during sport performance. Sport participants and athletes should drink enough water during exercise to prevent a deficit of more than 2 % to 3 % of body mass. They should aim to maintain a level of hydration that is optimal for performance and safety.

9.6.5 Each kilogram of body weight loss after training is equivalent to one litre of sweat loss (or fluid deficit). Using this equation, athletes should measure their pre- and post-training body mass during routine training and develop an individual hydration plan for use during races and competition events. A fluid deficit of more than 2 % to 3 % of body mass may compromise physical performance.

9.6.6 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should provide sport participants and athletes with advice on the risks of over-hydration leading to water intoxication.

10 Event medical support plan

10.1 General

10.1.1 An event medical support plan establishes appropriate medical support services required in mass-participation amateur sport events, based on the event risk assessment. It is an essential support tool to good safety practices and planning. Any medical support plan should complement the event safety plan.

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10.1.2 Even with the best preventive measures in place, injuries and incidents may still happen. Still, sport safety should be a primary focus as it is the most effective measure for minimising exercise-related injuries and mortality.

10.1.3 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and trainers should have a good medical support plan to provide persons with injuries or medical conditions with timely and appropriate medical support, manage injured persons effectively to reduce the impact of their medical condition, and reduce the exercise-related morbidity and mortality.

The following principles should be observed during planning:

- (a) Pre-participation baseline health assessment;
- (b) Sport safety and health education;
- (c) Pre-event and event day risk assessment and mitigation;
- (d) Robust and well-resourced medical support plan;
- (e) Close coordination and effective execution of the medical support plan;
- (f) Post-event feedback and review.

10.1.4 The medical support plans should be clearly communicated to the working personnel for effective implementation. Emergency contact information should be made available to the participants, volunteers, spotters, officials and staff.

10.2 Event medical coverage

10.2.1 The event medical support plan should primarily cater to event participants. In large-scale events with large number of spectators and organising staff, the medical support plan should also cater to the possibility of casualties coming from spectators or event organisers.

The following should be taken into consideration when organising large scale events:

- (a) Event medical support plans for large-scale events should be formalised and communicated to associated agencies, receiving hospitals who may have to manage security, medical evacuation and mass casualty situations.
- (b) Large-scale events should conduct medical support table-top and field exercises to ensure that the medical support plans including the evacuation routes and communication networks and channels are feasible.
- (c) Large-scale events should also conduct route reconnaissance to evaluate safety risks for runners, swimmers, cyclists, etc. during route planning.

10.2.2 Event medical support plans for mass-participation amateur sport should involve the management of a large number of participants and have unique requirements.

10.3 Risk assessment and categorisation

10.3.1 Sport events vary widely in terms of risk assessment and categorisation, based on the type of sport, level of competition, age group and skill level of participants, and other factors. The matrix assessment approach categorises sporting events into low, medium and high-risk categories (labelled as categories A, B and C respectively in Table 1). Table 2 categorises event risk based on

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cardiovascular risk and contact/collision risk, which is recommended with close reference from international publications and reports.

10.3.2 For training sessions and competitions in categories B and C (refer to Table 2), sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should conduct a risk assessment and have a medical support plan.

10.4 Resource management

10.4.1 As sport events vary widely with specific needs, it is not useful to have a one-size-fits-all guideline on the appropriate resourcing for sport events. Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should base their resource for training and competition built on each of the three risk categories within the risk assessment matrix as shown in Table 5.

Table 5 – Event medical coverage personnel and facilities

Risk category	Training or competition	Resource guidelines					
		First aider, CPR and AED trained official present (e.g. coach)	Dedicated first-aider or paramedic on-site	Doctor on-site	Nearest hospital informed	Ambulance on standby	Medical tent/post/centre on-site ¹
A – Low	Training	No	No	No	No	No	No
	Competition	No	No	No	No	No	No
B – Medium	Training	Yes	No	No	No	No	No
	Competition	Yes	Yes	No	Yes	No	No
C – High	Training	Yes	No	No	No	No	No
	Competition	Yes	Yes	Preferred	Yes	Preferred ¹	Yes, if >100 participants

¹ Refer to 10.4.4 for more information.

10.4.2 While the personnel have been identified by their vocational lines in Table 5, others with the same skill should also perform that role. Any person, beyond a first aider, such as a teacher, a volunteer or a bystander with the skills should identify any persons in distress and render first aid and CPR, if necessary.

10.4.3 For events that are dynamic/mobile or with a large field-of-play, sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers, coaches and instructors should adapt guidelines in Table 5. Resource guidelines in Table 5 applies primarily to static events, thus for events such as open-water kayaking, cycling and long-distance running where the medical support plan for training and competition should take into consideration the timely access of medical support and evacuation of casualties. For such events, the chief safety officer or event medical director should adapt the guidelines when drawing up the safety and medical support plan.

10.4.4 For category C high-risk competitive events, there should be timely access to medical evacuation. This should be provided using any emergency medical services for timely access to care, taking into consideration, heavy traffic, inaccessible routes, terrain etc. that may increase the time to access for the emergency medical services. If the event medical director assesses that the response time is insufficient, or the nature of the event makes it challenging for evacuation, then plans for forward

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deployed ambulance coverage should be put in place. In addition, the event medical director should consider that managing a pre-planned event with known cardiovascular risks would create higher expectations for a more responsive on-site medical support system to mitigate these additional risks.

10.5 Manpower and training

10.5.1 A safety officer should be assigned by the event organiser or person-in-charge to take charge of the event safety and medical coverage. Medical support staff should be briefed on the medical support plans including the use of medical equipment, activation procedures, evacuation routes and communication channels.

10.5.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should consider the appointment of an event medical director to be closely involved and responsible for the planning and execution of the medical support plan.

This applies to events that involve the following:

- (a) Mass participation;
- (b) Simultaneous multiple events (e.g. Youth Olympic Games); and
- (c) A large field of play (e.g. road races, triathlons).

10.5.3 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should review the experience and qualifications of the appointed safety and medical personnel to ensure that they are able to perform their roles effectively. Different levels of personnel competencies are required for different risk categories and type of events (i.e. training or competition) as in Table 5.

10.5.4 Safety and medical personnel shall be certified, and certifications shall be kept current. These should be based on national standards and recognised accreditation programmes (e.g. basic cardiac life support training and certification with training centres accredited by the Singapore Resuscitation and First Aid Council).

10.5.5 Sufficient numbers of trained safety personnel and first responders should be deployed to ensure that time required for the first responder to attend to a participant in distress is 3 min or less.

10.5.6 For specialised events such as water sport (e.g. swimming pool and open water) or disability sport, it is essential that personnel involved in event coverage shall be adequately trained in open water rescue techniques. Necessary skills shall include surveillance techniques, rescue and lifesaving skills for water sport, or being equipped to work with and rescue persons with particular disabilities, etc. For recurrent events, the event medical plan shall be reviewed along with the safety plan for continued improvement in subsequent events.

10.6 Medical equipment

10.6.1 Specialised personnel shall have full and rapid access to their respective equipment required. For instance, first aiders shall have access to an adequately equipped first aid box and medical support teams shall have well-equipped and stocked medical items. The medical items carried by medical teams for field shall contain resuscitation equipment (e.g. airway equipment, drip sets and fluid for resuscitation, bandages and dressings, etc.).

10.6.2 Commonly required medical equipment includes the following:

- (a) First aid box/kit;
- (b) Facemask for CPR;

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- (c) AED;
- (d) Oxygen (for water sport);
- (e) Splints and cervical collars;
- (f) Medical items for medical support teams;
- (g) Stretcher (regular or wheeled); and
- (h) Evacuation vehicle.

10.6.3 Equipment should be visible and accessible, kept well-stocked and well-maintained.

10.6.4 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should ensure that medical teams have prompt access to casualties.

10.6.5 Physical and administrative barriers should be addressed during the planning phase of the event. The safety of participants should be the prime consideration for event organisers and medical support personnel.

10.6.6 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers shall have one or more well-maintained AEDs that are easily accessible and located by users. The goal is to achieve the response time of 3 min or less – the duration from the time of cardiac arrest to defibrillation. Therefore, the absolute number of AEDs in each facility is dependent on the layout and size of the facility. Sport facilities shall store AEDs in a transparent cabinet that is highly visible/prominent and accessible, ensuring clear signage for users to locate the nearest AED. An alarm system with indicator lights alerts the public when the AED is removed. All facilities where exercise, training, and competition take place shall have public access defibrillators. Sport organisations that require additional mobile AEDs for event medical coverage may loan units from vendors.

10.6.7 Evacuation and safety vehicles should be adequately equipped, operated and located to enable effective medical assistance. For instance, safety and rescue boats should facilitate the transfer of casualties in and out of the boat and be given adequate space to carry out resuscitation. They should be appropriately equipped, competently manned and driven (to minimise the risk of propeller injuries to the victim).

10.7 Communication and coordination

10.7.1 All key event management and event medical personnel should have a clear communication plan covering safety and medical requirements.

This should cover the following:

- (a) First aid station locations;
- (b) Signage;
- (c) Ingress and egress routes for evacuation vehicles;
- (d) Types of medical vehicles to be used: bikes, buggy, ambulances, etc;
- (e) Medical protocols and procedures;
- (f) Inclement weather and haze;
- (g) Activation of medical team and evacuation process; and

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(h) Zone/sector coverage of lifesaving and rescue teams for water sport.

10.7.2 To provide timely lifesaving first aid to a participant or spectator, the following procedures should be clearly communicated:

- (a) Identification of victim;
- (b) Call for help;
- (c) Dispatch of medical team;
- (d) Transportation of medical team and equipment to victim or vice versa; and
- (d) Level of on-scene care.

10.8 Mass-participation amateur sport and events

10.8.1 Local medical coverage should take into consideration the adverse effects of Singapore's climate and weather such as heat, humidity and lightning. The local climatic conditions in Singapore may require the greater use of facilities such as additional water points, water sprays and cooling systems to minimise the likelihood of adverse effects (e.g. heat exhaustion and heatstroke).

10.9 Post-event feedback and review

10.9.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should gather post-event feedback on the safety management and medical support plan to improve the organisation of future events.

10.9.2 For events endorsed by the local authority for a particular sport, sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should submit their post-event report to such authority.

11 Exercise and training facilities

11.1 General

11.1.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions, coaches, instructors and event organisers should promote and advise all users to use the pre-participation questionnaire as well as orientating and guiding them in the safe usage of the facility. In facilities with free uncontrolled usage of its equipment (e.g. fitness parks or unsupervised hotel fitness centres), pre-participation questionnaires should be easily accessible, and users are made aware of its presence. This could be done through websites or by means of signs at the facility.

11.1.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions, coaches, instructors and event organisers should adopt risk management and established emergency policies, and professionally operate a facility. These may mitigate unnecessary risks and promote a positive experience for users.

11.1.3 Sport organisations, associations, clubs, facility owners and operators, educational institutions, coaches, instructors and event organisers shall ensure staff and independent contractors are professionally equipped and certified, as well as familiarise with the safety practices of the facility.

11.1.4 For design and management of aquatic facilities, SS 556 shall apply.

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11.2 Pre-participation questionnaires

11.2.1 Sport participants and users should adhere to proper training principles and guidelines to pursue physical activity in a safe manner.

11.2.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions, coaches, instructors and event organisers should advise participants and users with existing pre-participation guidelines and seek medical clearance from a qualified healthcare provider, if required.

11.3 Orientation and supervision

Sport organisations, associations, clubs, facility owners and operators, educational institutions, coaches, instructors and event organisers should provide assistance and guidance in relation to physical activity to facility members and users engaged in exercise and training programmes.

11.4 Risk management and emergency policies

11.4.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should have a well-documented emergency response policies and procedures regularly reviewed. Physical rehearsals/drills should be conducted twice annually to ensure that employees are equipped to respond to basic first aid situations and emergency events in an appropriate and timely manner.

11.4.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should conduct safety audits to mitigate injury-causing hazards, thus keeping employees, facility members and users safe.

11.4.3 Sport organisations, associations, clubs, facility owners and operators, educational institutions, event organisers, coaches and instructors should use waivers of liability and/or assumption of risk documents with facility members and users.

11.4.4 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should deploy AEDs within their facilities with the goal of being available to any casualty within 3 min of call. A notice highlighting the availability of an AED at the facility (or nearby with the directions) should be prominently displayed near entry to the facility. AED signage(s) should follow the Singapore resuscitation and first aid council (SRFAC) guidelines. In larger facilities, directional arrows are helpful when there is no direct line-of-sight to the AED location.

11.4.5 Staffed facilities should have at least one staff member on duty during operating hours. At least one staff member on duty should be currently trained and certified in the delivery of CPR and in the use of an AED.

11.4.6 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers shall provide employees with the opportunity to receive training and certification in first aid, CPR and AED followed by refresher courses and on-the-job training. Employees should undergo recertification in first aid, CPR and AED once their certification expires.

11.4.7 Unstaffed facilities should have a public access defibrillation programme for members and users (or an external emergency responder) to use in response to an emergency. This should enable a response in 3 min or less (from the time of call to defibrillation) and should be part of a facility's documented emergency response policies and procedures.

11.4.8 Refer to Table 6 on staffing and emergency equipment in exercise and training facilities.

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Table 6 – Staffing and emergency equipment in exercise and training facilities

Facility characteristic	Level 1	Level 2	Level 3	Level 4	Level 5
Type of facility	Unsupervised exercise room	Single exercise leader	Fitness centre for healthy clients	Fitness centre serving clinical populations	Medically supervised clinical exercise programme
Personnel	None	Exercise leader Recommended: Medical liaison	General manager Fitness instructor Exercise leader Recommended: Medical liaison	General manager Exercise specialist Fitness instructor Medical liaison	General manager Exercise specialist Fitness instructor Medical liaison
Emergency plan	Present	Present	Present	Present	Present
Emergency equipment	Telephone in room/ parks Signs Encouraged: PAD plan with AED as part of the composite PAD plan in the host facility (e.g. hotel, commercial building, apartment complex, park)	Telephone in room Signs Encouraged: blood pressure kit PAD plan with AED	Telephone in room Signs Encouraged: blood pressure kit, stethoscope PAD plan with AED	Telephone in room Signs Blood pressure kit Stethoscope Strongly recommended : PAD plan with AED	Telephone in room Signs Blood pressure kit Stethoscope Oxygen Crash cart Defibrillator
Key AED – Automated external defibrillator PAD Public access defibrillator					

NOTE – The level of the facility is not based on the size of facility.

11.5 Professional staff and independent contractors

11.5.1 Staff that administer physical activity programmes or have supervisory responsibilities should meet professional standards (e.g. certification) in their training and experience according to their designation.

A list of roles and training/certification for fitness centre personnel is shown in Table 7.

11.5.2 Professionals engaged in pre-activity screening or prescribing, instructing, monitoring, or supervising of physical activity programmes should have current AED and CPR certification from a local organisation accredited by the national accreditation agency.

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Table 7 – Roles and training/certification for fitness centre personnel

Designation	Roles and training/certification
General manager/ Executive director	Overall management of facility, design and delivery of exercise programmes
Medical liaison	Review of medical emergency plans and incident reports, witnessing and critique of emergency drills. Levels 2 and 3 facilities: Physician or registered nurse trained in advanced cardiac life support Levels 4 and 5 facilities: Physician
Fitness director	Design and management of exercise and fitness programmes, training and supervision of staff core requirement Bachelor's degree in exercise science (or similar/related field e.g. physical education, exercise & sport science) with more than 5 years of experience in the fitness industry, of which 3 years are in supervisory appointments, or Diploma in exercise science (or similar/related field e.g. physical education, exercise & sport science) with more than 8 years of experience in the fitness industry, of which 5 years are in supervisory appointments Valid certification in standard first aid with CPR and AED Fitness trainer certifications and/or relevant and recognised qualifications
Fitness professionals (including professional exercise leaders, personal trainers, etc.)	Provide instruction and leadership in professional exercise programmes Fitness trainer certifications and/or relevant qualifications that are recognised Valid certification in standard first aid with CPR and AED Prior supervised internship or experience in health/fitness industry

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11.6 Facility operating practices

11.6.1 The following facility operating practices should apply to facility owners and operators:

- (a) To have a manual or digital operations system that monitors the presence and identity of individuals that enter the facility or participate in activities, programmes and services (e.g. members, users).
- (b) To ensure that there is sufficient space (both physical activity space and non-activity spaces) to accommodate user-demand.
- (c) To have a technical monitoring system and AED in place for facilities such as sauna, steam room, or whirlpool to ensure that the areas are maintained at the proper temperature and humidity level. Appropriate warning systems and signage should be in place to notify members and users of risks related to the use of these facilities. This includes warning users about unsafe changes in temperature and humidity.
- (d) To display written policies visibly, for instance, restroom practices, age requirements or entry limits for children and the need for parental/guardian presence on-site.
- (e) To maintain background noise levels (below 70 decibels and never above 90 decibels).
- (f) To ensure that all floor surfaces in physical activity areas for the proper level of absorption and slip resistance to minimise the risk of fall-related injuries.

11.7 Facility equipment

11.7.1 Sport organisations, associations, clubs, facility owners and operators, educational institutions shall carry out installation of sport playing equipment according to safe installation instructions from manufacturers manual as improper installation may result in serious injury or death. Checks on the stability of sport structures including supporting structures should be conducted on a regular basis as it can deteriorate over time especially due to high usage. There should be a preventive maintenance programme for fitness equipment, including updated documentation indicating when the scheduled work was performed.

11.7.2 The following should apply to sport organisations, associations, clubs, facility owners and operators, educational institutions:

- (a) To display visually prominent notices on danger and warnings of hazards, and proper caution.
- (b) To display visually prominent signage indicating the location of any rescue/lifesaving equipment, such as AED and first aid kits.
- (c) To post the appropriate emergency and safety signage pertaining to fire and related emergency situations, such as fire extinguishers and hose reels.
- (d) To display advisories prominently to remind participants of unsafe behaviours, not to horseplay or perform any high-risk activities without proper supervision.

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12 Surveillance and evaluation

12.1 General

12.1.1 Injury surveillance is integral to improving sport safety.

12.1.2 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should conduct debriefings and post-event reviews of sport event.

12.1.3 Sport organisations, associations, clubs, facility owners and operators, educational institutions and event organisers should conduct injury surveillance, capture data on sporting injuries, sudden deaths and serious adverse events for trend analysis.

12.2 Sporting injuries data

The following should apply to sport organisations, associations, club, facility owners and operators, educational institutions and event organisers:

- (a) To use a minimum data set defined by sport organisations to allow easier compilation of standardised data (Refer to the sport injury reporting form in Annex B). Stakeholders and organisations shall incorporate this data set into their own reporting formats.
- (b) To compile data on participants with medical problems detected during screening (using their own reporting system/protocol, including the minimum data set) or those injured during sport events (using the sport injury reporting form in Annex B) and document a follow-up plan. Allow the above data to be easily accessible by relevant parties (e.g. medical personnel) while protecting medical confidentiality. The compilation and evaluation of such data in each sport allow for more accurate risk stratification and also provide information on the adequacy of event safety plans.
- (c) To conduct its own injury surveillance among its sport participants and athletes. This enables comparisons with international benchmarks and the systematic improvement of sport safety.
- (d) Where applicable, sport organisations, associations, clubs, educational institutions, facility owners and operators and event organisers should establish their own sport-specific medical committee (or at least appointing a medical advisor) to help comply with the above recommendations, including periodically reviewing data collected on sport-related injuries/adverse events, and managing the data. For a start, the information could be quite basic with minimal data requirements and the information could be refined/increased as the process matures.
- (e) To conduct debriefings and post-event reviews of sport events. They should promote sharing of essential points amongst organisations, and invite feedback from members of the public and participants. Improvements can be publicised to raise public awareness (via sport organisations/ event websites, etc.).
- (f) To ensure their surveillance and evaluation systems contain a follow-up and tracking element to prevent incidents of similar nature in the future.
- (g) To review their sport injury registry to provide the following functions:
 - Verify compliance with rules and guidelines;
 - Develop guidelines on safe conduct of activities, spread best practices, and promote general awareness of sport safety; and
 - Capture data on sporting injuries, sudden death and adverse events, to provide an objective basis for the design and continual improvement of effective interventions.

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Annex A (informative)

Checklist of intrinsic risk factors for exertional heatstroke

Are you at risk for exertional heatstroke?		
1	Are you physically ready for training and competition?	Athletes need to match their exercise intensity with their fitness level. Novice athletes with poor physical fitness tend to outpace themselves during competition. An overload in exercise intensity is a key factor contributing to exertional heatstroke.
2	Have you acclimatised to the climate?	In Singapore's context, it is important for visiting athletes to have adequate acclimatisation to the heat and local climate.
3	Have you been ill recently?	Heatstroke risk can be increased by disturbances to the immune system (e.g. from a recent bout of illness or sub-clinical infection).
4	Are you taking any medications?	Stimulants, antihistamines, diuretics and other common medications can impair the body's ability to mount an effective thermoregulatory response during exercise in the heat. Athletes taking medication for recent illness should be advised against participation in view of the dual risks medication and recent illness pose.
5	Do you tend to push your body hard?	An athlete's high level of motivation is one of the most consistent hallmarks of exertional heatstroke. Signals which urge the body to slow down are blocked out when athletes are knowingly pushing their limits to reach the highest levels of performance.
6	Are you at risk for heat injury?	High body mass: High body mass athletes expend metabolic energy at higher absolute rates than their lean counterparts, placing themselves at higher risk of heat injury. Children and youth: Children and youth may be unable to assess and mitigate risks of training and competing in the heat. The role of age and sex in heat injury risk is a subject of debate. Thermoregulatory responses in children were once thought to be drastically different from those in adults, although a recent review described no differences.

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Annex B (informative)

Sport injury reporting form

Name of injured person: _____

Gender: M / F

Age: _____

Contact. No.: _____

Player/Technical official/Coach/ Spectator

Date: ____/____/____

Time: ____:____

Venue: _____

Sport: _____

Training/Competition

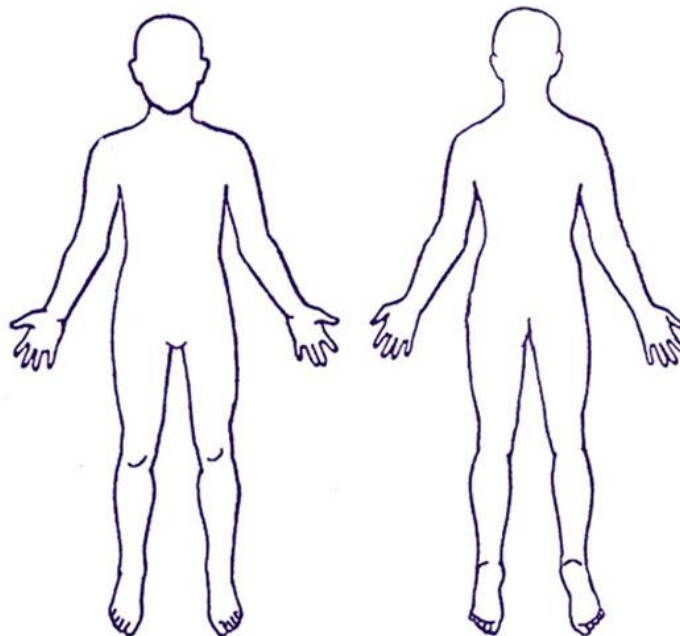
New injury/Recurrent injury/Pre-existing illness

Condition of injured person on treatment:

Conscious/Semi-conscious/Unconscious

Body region involved

- ☐ Head and Neck
- ☐ Torso
- ☐ Abdomen
- ☐ Groin/Pelvis
- ☐ Upper limb
- ☐ Lower limb
- ☐ Others: _____



FRONT

BACK

Nature of Injury

(a) Abrasion/Blisters	Laceration Bruise/Contusion
(b) Swelling	Sprains/Ligament injury Strain muscle injury
(c) Fracture	Dislocation/Subluxation
(d) Concussion/Head injury	

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(e) Facial Injury – Eye, dental trauma
(f) Cardiac contusion/angina/arrest
(g) Respiratory distress
(h) Abdominal cramp/vomiting
(i) Heart-related illness
(j) Submersion/drowning
(k) Others:-

State if injured person is bleeding

Slight/Minor/Heavy/Require stitches

Cause of injury

- ☐ Struck/Collision by another player
- ☐ Struck/Collision by ball or object of play
- ☐ Fall on same level or while jumping/poor landing
- ☐ Overexertion/Overuse
- ☐ Heat-related
- ☐ Others: _____

Diagnosis: _____

Description of incident:

Site of incident

- ☐ Arena
- ☐ Rest area
- ☐ Poolside/Water play area
- ☐ Diving board/Platform
- ☐ Shallow water – Depth: ___m
- ☐ Deep water – Depth: ___m
- ☐ Distance from starting point:___km
- ☐ Others: _____

Protective equipment used? Y / N _____

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Initial treatment

- Rest, ice, compression, elevation
- Dressing
- Stretching
- Splint/Sling
- CPR required? No. of cycles
- AED required? No. of “shocks” delivered
- Others: _____

Outcome

- Return to play
- Rest at side
- Referred to specialist outpatient clinic
- Referred to hospital A&E
- Conveyed to hospital A&E by ambulance

Treated by

- Paramedic
- First aider
- Sport trainer
- Others: _____

Name of medical personnel: _____

Signature: _____

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Summary of sport injury report

(To be compiled at the end of the event)

Date: ____/____/____

Reporting time in: ____:____

Reporting time out: ____:____

Venue: _____

Sport: _____

Number of participants: _____

Type of injury	Number of incidents	Treated on-site	Conveyed to hospital
A. Abrasion/Blisters/Laceration/Bruise/Contusion			
B. Swelling/Sprains/Ligament injury/Strain muscle injury			
C. Fracture/Dislocation/Subluxation			
D. Concussion/Head injury			
E. Facial Injury – Eye, Dental trauma			
F. Cardiac Contusion/Angina/Arrest			
G. Respiratory distress			
H. Abdominal cramp/Vomiting			
I. Heat-related Illness			
J. Submersion / Drowning			
K. Others: _____			

Cause of Injury

____ Struck/Collision by another player

____ Struck/Collision by ball or object of play

____ Fall on same level or while jumping/poor landing

____ Overexertion/Overuse

____ Heat-related

____ Others: _____

Outcome

Total no. of referred to hospital A&E: _____

Total no. conveyed to hospital A&E by ambulance: _____

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Prepared by:

Name of medical personnel: _____

Contact No.: _____

Signature: _____

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- [3] The Singapore Resuscitation and First Aid Council (SRFAC) guidelines
- [4] The United Kingdom Athletics' (UKA) Road race medical services good practice guide

Resources from the Sport Singapore Website

- [5] Risk assessment and management systems: <https://www.sportsingapore.gov.sg/sports-education/sports-safety/safety-resources-and-useful-links>
- [6] Risk assessment methodology/template: https://www.sportsingapore.gov.sg/-/media/SSC/Corporate/Files/Sports-Education/Sports-Safety/Safety-Resources-and-Useful%20Links/Risk_Management_In_Sport.pdf?la=en&hash=87D6B3A4F0F795295DE3DF449F2B40F60E4E02B0
- [7] Coach's Code of Ethics: <https://www.sportsingapore.gov.sg/athletes-coaches/coaches-corner/code-of-ethics>.
- [8] Example of a pre-participation questionnaire: <https://www.sportsingapore.gov.sg/Sports-Education/Sports-Safety/Safety-Resources-and-Useful-Links>.

NOTE – Hyperlinks are valid and current at the time of publication of this standard.

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Changi General Hospital
Health Promotion Board
KK Women's and Children's Hospital
Ministry of Home Affairs
Nanyang Technological University
National University of Singapore
Ng Teng Fong General Hospital
SAFRA National Service Association
Singapore Rugby Union
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